PR GUARDIAN

Four parallel IBM Bob specialists.
One severity-ranked report.

SECURITY PERFORMANCE TESTS ARCHITECTURE

12/12 seeded defects caught
~1 Bobcoin per review



Built with IBM Bob IDE 2.0 · lablab.ai IBM Bob 2.0 Hackathon · Akobir Nuraliyev

Code review is where shipping slows down — and where bugs slip through

Days of waiting

A PR often waits longer for review than it took to write. Work piles up behind it.

One human, four jobs

The reviewer must be a security engineer, a performance engineer, a test engineer and the keeper of the team's conventions — at once.

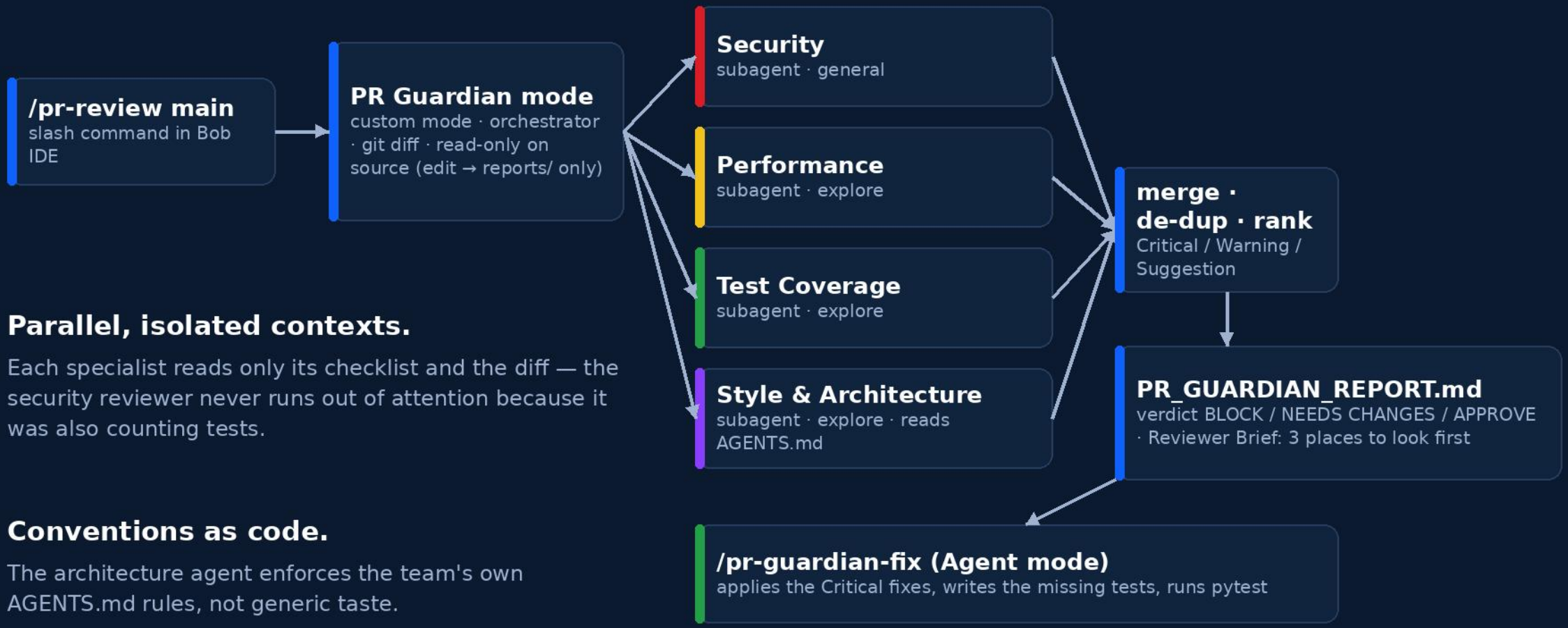
Fatigue = misses

At 6 p.m., between meetings, a tired reviewer skims. SQL injections, missing auth checks and N+1 queries reach production.

Teams either block everything on review — or accept the risk.

Neither is acceptable. The reviewer needs help before they open the diff.

/pr-review main → four Bob specialists, one ranked report



Parallel, isolated contexts.

Each specialist reads only its checklist and the diff — the security reviewer never runs out of attention because it was also counting tests.

Conventions as code.

The architecture agent enforces the team's own AGENTS.md rules, not generic taste.

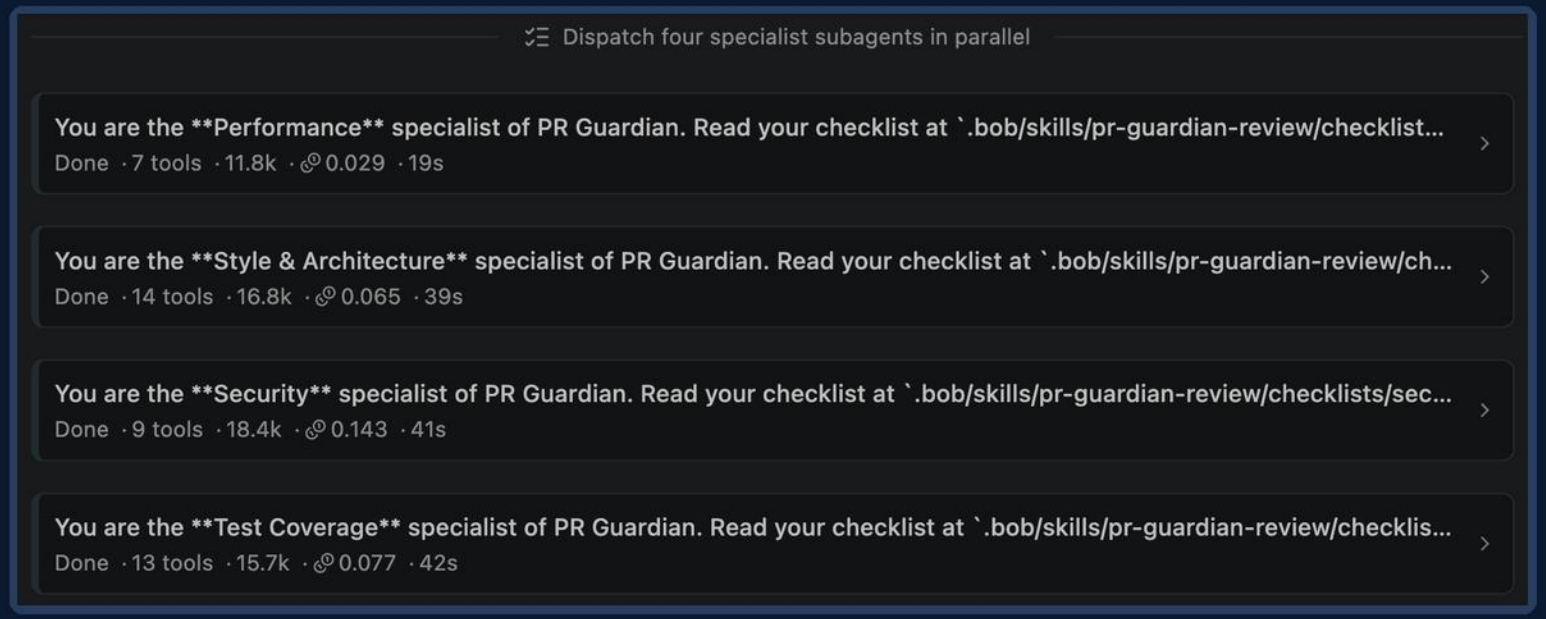
Built entirely on IBM Bob 2.0 — Bob is the runtime, not a helper

Everything is configuration in .bob/ — copy the folder into any repository

Custom mode	.bob/custom_modes.yaml — role, whenToUse, tool groups; edit restricted by fileRegex to reports/; allowedSubagents
Subagents	4 spawned in one turn: Security on general, three on explore — parallel panel, isolated contexts
Skill	.bob/skills/pr-guardian-review/ — protocol + finding schema + report template + 4 checklists
Mode rules	.bob/rules-pr-guardian/ — condensed workflow, severity policy (incl. the tie-breaker Bob proposed)
Slash commands	.bob/commands/ — /pr-review $1 and /pr-guardian-fix
AGENTS.md + .bobignore	repo conventions for the architecture agent; benchmark answer key hidden from Bob
Agent mode	remediation step: fixes that follow AGENTS.md, generated tests, pytest green
Task History export	bob_sessions/ — consumption summaries + exported task history for judging

Live run: four specialists, one turn, in parallel

feature/search-export-report vs main — 3 files, 102 lines



42 s
wall-clock for all four specialists

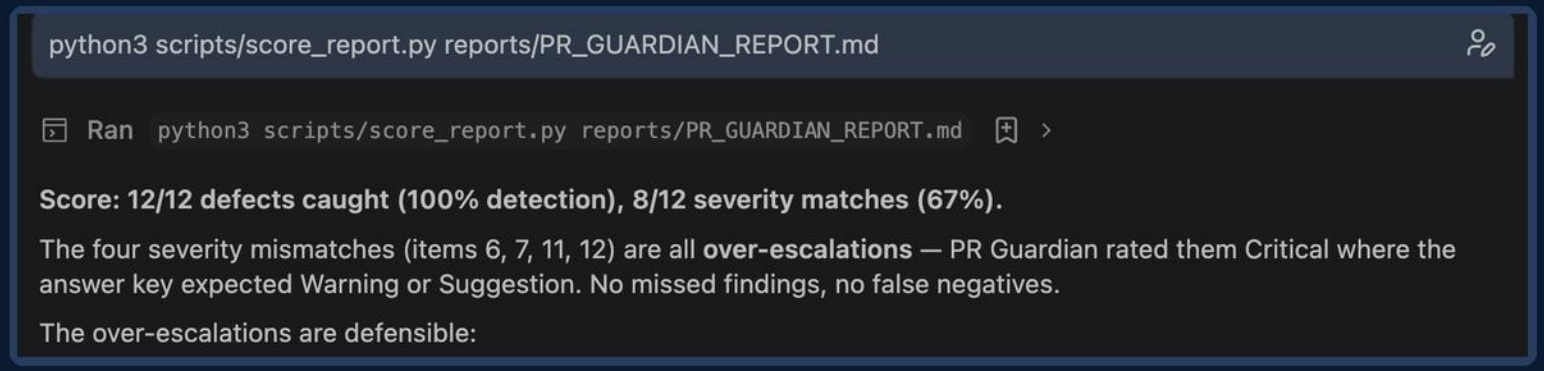
141 s
of specialist work, run concurrently

0.31
Bobcoin for the four subagents

0.96
Bobcoin for the whole review task

Seeded benchmark: 12 / 12 caught, all four Criticals

12 planted defects · answer key hidden from Bob via .bobignore · scored by scripts/score_report.py



Caught

- SQL injection (f-string query)
- hard-coded webhook secret
- unauthenticated admin endpoint
- IDOR on CSV export
- N+1 queries, O(n^2) loop, two leaks
- zero tests for 4 endpoints
- layering + naming violations

12/12
defects detected

4/4
Critical security flaws

8/12
exact severity; 4 over-escalated (safe side)

BLOCK
verdict — before a human spent a minute

Business value and what makes it different

Before → After

Cold diff at 6 p.m.
→ **Ranked Reviewer Brief: the 3 lines to read first**

Critical issues wait days
→ **Critical issues surface in under a minute**

One tired generalist
→ **Four fresh specialists + a human who decides**

Conventions in people's heads
→ **Conventions in AGENTS.md, enforced on every PR**

New tool, new CI, new infra
→ **Copy one folder into the repo — done**

vs. Bob's built-in /review

- Specialists with isolated attention instead of one generalist pass
- Repo-convention awareness through AGENTS.md
- Severity policy is a file the team edits — Bob itself proposed the first refinement
- Least-privilege orchestrator: can only write reports/
- Reproducible seeded benchmark to prove recall and cost
- Cost: about one Bobcoin per review — cheaper than one missed injection

What's next — and where to look

Roadmap

- Bob Shell headless run in a GitHub Action: post the report as a PR comment
- Bob workflow version: deterministic diff/score steps + parallel AI steps
- Per-repo severity policy and per-language checklists
- Reviewer Brief pushed to chat so the human starts warm

Links

Repository
github.com/AkobirNuraliyev11/pr-guardian

Pull request under review
.../pr-guardian/pull/1

Evidence
bob_sessions/ — Bob IDE task summaries + exports

Statements
docs/STATEMENTS.md

Try it
copy .bob/ into your repo → /pr-review main

PR Guardian — four Bob specialists, one report, before a tired reviewer opens the PR.

Akobir Nuraliyev · solo · built with IBM Bob IDE 2.0